MODULE 48 · DOMAIN 5 - OPERATIONAL EXCELLENCE AND AUTOMATION

Operational Automation and Resilience Patterns

Let events start the fix, let the code survive its dependencies, let capacity arrive before the load, and let unhealthy parts replace themselves.

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Design so that the system **contains** a failure and **repairs** itself, and a person only decides what the automation cannot.

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Resilience patterns in code

PATTERN	PREVENTS	ON AWS
Timeout	Waiting forever on a slow call	SDK and client settings
Retry, backoff, jitter	Retry storms in step	AWS SDK retry modes
Circuit breaker	Hammering a failing dependency	Library or Step Functions logic
Bulkhead	One workload starving another	Reserved concurrency, separate pools
Rate limit	Overload from one caller	API Gateway usage plans, WAF rate rules
Idempotency	Duplicate effects from retries	Idempotency keys, Powertools

ON THE EXAM

Retries without backoff, jitter and idempotency make an outage worse.

Scaling policies

POLICY	HOW IT DECIDES	BEST FOR
Target tracking	Hold a metric at a target	Most workloads, the default
Step scaling	Act by alarm breach size	Custom, tiered responses
Scheduled	Change capacity at set times	Known one-off events
Predictive	Forecast from history, launch early	Daily or weekly cycles, slow start

ON THE EXAM

Regular cycle, slow warm-up: add predictive scaling alongside target tracking.

Self-healing

COMPUTE	MECHANISM
EC2 Auto Scaling	ELB health checks replace unresponsive instances
EC2 single instance	Automatic recovery on hardware failure
ECS	Unhealthy tasks stopped and replaced
Lambda, asynchronous	Retries, then on-failure destination or DLQ
Lambda, SQS source	Failed messages to a dead-letter queue

ON THE EXAM

EC2 status checks pass while the application hangs: use ELB health checks.

Event-driven operations

01
Event
Alarm, health event, Config change

02
EventBridge
Rule matches, or Scheduler fires

03
Runbook
SSM Automation with approvals

04
Record
OpsItem, notification, audit

ON THE EXAM

Incident Manager and Change Manager closed to new customers on 7 Nov 2025.

Reading the scenario

WHEN THE QUESTION SAYS	IT IS TESTING
Slow dependency, failure cascades	Circuit breaker and timeouts
Retries arrive in waves	Exponential backoff with jitter
One workload starves another	Bulkhead, reserved concurrency
Daily peak, scaling too late	Predictive scaling
App hangs, instance not replaced	ELB health check type
Quota limits hit across accounts	Service Quotas automatic management

ON THE EXAM

Contain, repair, scale ahead, automate the fix.

> Hollins Grocers' checkout service calls a third-party payment provider. When the provider slowed to 30-second responses, checkout threads filled up waiting, and the whole site stopped responding.
>
> The team wants a provider problem to degrade checkout, not take down the site.

What should the solutions architect recommend?

A Increase the client timeout to 60 seconds

B Short timeouts and a circuit breaker that stops calling the provider when it fails, returning a fallback such as queueing the payment

C Retry each failed call five times immediately

D Move checkout to larger instances

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Longer timeout

Threads wait even longer.

B Timeouts and circuit breaker

Fails fast, stops calling, degrades gracefully.

C Immediate retries

Multiplies load on a struggling provider.

D Larger instances

More threads to fill; same outcome.

THE SIGNAL

Slow dependency spreading to the whole system: timeouts and a circuit breaker.

> After a brief DynamoDB throttling event, Hollins Grocers' 2,000 order workers all retried exactly one second later, causing a second, larger spike, then a third.
>
> Some orders were also recorded twice.

What should the solutions architect change?

A Remove retries entirely

B Retry every 100 milliseconds instead

C Use exponential backoff with jitter for retries, and make order writes idempotent with a client token

D Request a higher table throughput only

◆ PAUSE AND DECIDE

What should the solutions architect change?

A No retries

Turns brief throttles into failed orders.

B Faster fixed retries

Waves arrive sooner, still in step.

C Backoff, jitter, idempotency

Spreads retries; duplicates have no effect.

D More throughput only

Helps capacity; duplicates and waves remain.

THE SIGNAL

Retries arriving in waves, plus duplicates: backoff with jitter, and idempotency.

> Hollins Grocers runs checkout and a nightly catalogue import as Lambda functions in the same account. Last night the import scaled to the account's concurrency limit, and checkout requests were throttled.
>
> Checkout must always have capacity, whatever the import does.

What should the solutions architect configure?

A Provisioned concurrency on the import function

B Move checkout to EC2

C Raise the account concurrency limit only

D Reserved concurrency for checkout, and a maximum concurrency on the import's SQS event source

◆ PAUSE AND DECIDE

What should the solutions architect configure?

A Provisioned on import

Pre-warms the wrong function; no limit.

B Move to EC2

Large change for an isolation problem.

C Higher limit only

The import can still grow into it.

D Reserved and capped

Checkout keeps its share; import is bounded.

THE SIGNAL

One workload starving another: a bulkhead, with reserved and capped concurrency.

> Hollins Grocers' web tier in an Auto Scaling group sees traffic triple every weekday between 08:30 and 09:00. Instances need eight minutes to warm their caches.
>
> Target tracking reacts, but customers see slow pages for the first ten minutes every morning.

What should the solutions architect add?

A. Predictive scaling, alongside the existing target tracking policy

B. Step scaling with larger steps

C. A lower target tracking threshold

D. Double the minimum capacity all day

◆ PAUSE AND DECIDE

What should the solutions architect add?

A Predictive scaling
Launches capacity before the daily peak.

B Bigger steps
Still reacts after load arrives.

C Lower threshold
Earlier reaction, still too late for warm-up.

D Double minimum
Pays for peak capacity all day.

THE SIGNAL

Regular daily peak plus slow warm-up: predictive scaling.

> Hollins Grocers' application on EC2 sometimes deadlocks. The instance stays up and passes EC2 status checks, but stops answering requests. The Application Load Balancer marks it unhealthy, but the Auto Scaling group never replaces it.
>
> The team wants such instances replaced automatically.

What should the solutions architect change?

A Enable EC2 automatic recovery

B A CloudWatch alarm that reboots the instance on high CPU

C Set the Auto Scaling group health check type to ELB, with a suitable grace period

D Use a larger instance type

◆ PAUSE AND DECIDE

What should the solutions architect change?

A **Auto recovery**

Acts on hardware and system failures only.

B **Reboot on CPU**

A deadlock may not raise CPU.

C **ELB health checks**

The group then replaces instances the ALB finds unhealthy.

D **Larger instance**

Deadlocks are not about size.

THE SIGNAL

Load balancer sees it unhealthy, group does not replace it: ELB health check type.

> Hollins Grocers has 200 accounts. Launches keep failing when a team hits a service quota, such as Elastic IP addresses or Lambda concurrency, and a quota request is raised by hand.
>
> The platform team wants warnings as usage nears a quota, and quotas adjusted automatically based on usage.

What should the solutions architect use?

A. A weekly review of Trusted Advisor service limit checks

B. Service Quotas automatic quota management, with notifications

C. A Lambda function that requests the maximum for every quota

D. Ask teams to raise requests earlier

◆ PAUSE AND DECIDE

What should the solutions architect use?

A Weekly review

Manual and infrequent.

B Automatic quota management

Alerts near limits, adjusts by usage, no charge.

C Request maximum

Unjustified requests; removes useful guard rails.

D Earlier requests

Still manual.

THE SIGNAL

Warn near quotas and adjust them by usage: Service Quotas automatic management.

What to carry out of this module

- **Timeouts, circuit breakers, bulkheads** contain failures.
- **Backoff with jitter** plus **idempotency** make retries safe.
- **Predictive scaling** for regular peaks and slow warm-up.
- **ELB health checks** replace hung instances.
- **EventBridge to SSM Automation**; quotas managed automatically.

> ON THE EXAM
>
> Contain, repair, scale ahead, automate the fix.